

My Goal for this project is to dive deep into **exploratory data analysis**. **EDA** is one of the most important skills especially if I am transitioning into the industry, which is why understanding how to use the tools as a cohesive system to surface the questions that stakeholders are actually curious about is important for me.
Once I have got that next step is to translate those into meaningful insights, even when the question is very ambiguous.

The Question from Requirements Gathering:

My Question: “How did the sales look over the COVID years?”. After a little help from our ChatGPT friend, the Question becomes,

The operational manager asks **“How did total revenue dollars across all products performed during 2019 to 2022? Conduct initial analysis for managers in marketing , product and finance to understand high-level trends.”**

I am already done with cleaning of the data in here [gamezone-sales-data-cleaning](#).

For EDA I am going to use the SCAN framework and create an insights_log file where I can start noting down approaches and answers for each part of SCAN.

S Stakeholder Goal (What decision is this analysis supporting? What KPIs matter?)

C Columns and Coverage(What data do we have available? And how can I use it?)

A Aggregates and Anomalies (high level metrics, outliers, aggregates and unexpected patterns)

N Notable Segments (slice by category, time and other key dimensions to surface early insights).

Writing down all the answers to SCAN framework questions, Now that I have insights_log with my scan answers ready, I have a direction to keep coming back to.

For Stakeholder Goal Part -

There's no direction being made as of yet from my initial question or definition of this project. So going to focus on high level trends and patterns. And the most **important KPI is going to be the USD price.**

Now I am just going to be using the **USD price**, the **product Columns** and the **purchase time columns.**

So when I want to look at the **total USD price** across **months, products and avgs** I am going to use PIVOT tables.

For Columns and Coverage Part -

I am going to use USD_PRICE, PRODUCT_NAME, PURCHASE_TS columns.

For Aggregates and Anomalies Part -

When I want to look at total usd price across months, products, & avgs. Identify mins, maxs and any outliers. I am going to use Pivot tables to start piecing the total months, total products and the averages.

1. Pivot 1 for monthly usd price across the years, the question didn't say monthly exactly however I think monthly will show more insights than yearly.

Added **PURCHASE_MONTH** in Row and sum of **USD_PRICE** for the Columns, added extra filter of year so that I can see the yearly changes.

I can see we have made \$6.1 million and the total USD price range is around \$80k to \$550k.

Now I want to look into the product, I am going to add **PRODUCT_NAME_CLEANED** into the Pivot table.

With the month on the left and product on the top I can not only see how totals compare to each other across products but also how a product evolves over a time in terms of its sales.

Some Observation:

1. Total sales is \$6.1 million , per product it ranges from \$800 to almost \$2 million.
2. The gaming monitor and the Nintendo switch are performing much better than others whereas Razer Pro Gaming Headset is performing the worst.
3. Adding conditional formatting and data bars to see the seasonality proves my 1st two observations.

Sum of USD_PRICE	PURCHASE_YEAR	PURCHASE_MONTH	PRODUCT_NAME_CLEARED	Acer Nitro V Gaming Laptop	Dell Gaming Mouse	JB.L Quantum 100 Gaming Headset	Lenovo IdeaPad Gaming 3	Nintendo Switch	Razer Pro Gaming Headset	Sony PlayStation 5 Bundle	Grand Total
	2019	1	27in 4K gaming monitor	\$33,501.10	\$2,298.04	\$2,894.73	\$6,524.52	\$28,539.12		\$16,733.71	\$100,491.28
		2		\$26,409.22	\$1,463.76		\$1,810.60	\$7,548.12	\$25,381.51	\$17,775.89	\$80,389.10
		3		\$35,189.23	\$3,032.40		\$3,456.28	\$14,864.02	\$29,063.02	\$19,589.03	\$115,193.98
		4		\$48,754.08	\$2,126.96		\$3,389.99	\$10,771.88	\$37,500.86	\$18,885.26	\$121,408.83
		5		\$38,973.22			\$3,126.32	\$12,594.76	\$37,417.94	\$31,270.71	\$123,376.92
		6		\$37,462.19			\$3,086.08	\$13,328.81	\$34,324.02	\$18,696.19	\$108,450.52
		7		\$46,875.92	\$1,537.60		\$3,511.55	\$15,875.57	\$40,526.56	\$17,067.04	\$125,094.24
		8		\$40,005.11	\$648.22		\$3,234.25	\$15,184.91	\$47,284.50	\$23,194.88	\$129,552.30
		9		\$61,630.98	\$1,996.00		\$3,787.13	\$17,492.96	\$51,372.79	\$25,693.42	\$161,573.28
		10		\$50,115.27	\$1,574.29		\$2,730.79	\$10,980.86	\$35,971.09	\$20,750.08	\$122,162.32
		11		\$67,326.57	\$1,361.44		\$3,059.59	\$23,061.68	\$53,718.80	\$23,530.97	\$172,459.45
		12		\$72,965.34	\$745.04		\$4,250.07	\$16,867.59	\$63,574.77	\$22,366.51	\$180,769.32
	2020	1		\$41,477.16		\$49.98	\$1,574.33	\$13,840.22	\$25,296.73	\$27,707.21	\$109,935.63
		2		\$62,191.15	\$2,394.00	\$202.58	\$2,971.96	\$21,339.53	\$47,926.41	\$48,412.53	\$185,438.59
		3		\$96,828.17	\$2,249.81	\$485.53	\$4,294.14	\$40,423.75	\$70,043.39	\$119.98	\$86,033.26
		4		\$108,420.16		\$1,269.87	\$5,123.36	\$36,115.26	\$88,551.96	\$112.83	\$111,961.64
		5		\$114,238.47	\$4,371.76	\$1,261.40	\$5,141.88	\$38,958.11	\$53,834.56	\$119.98	\$87,587.50
		6		\$98,761.99	\$1,859.28	\$1,383.40	\$4,372.52	\$34,491.26	\$82,209.67	\$119.98	\$91,541.90
		7		\$100,514.34	\$3,192.00	\$1,535.00	\$4,478.32	\$36,098.31	\$93,331.86		\$64,286.78
		8		\$96,526.38	\$4,773.88	\$1,532.96	\$4,220.38	\$63,631.34	\$101,801.47	\$138.26	\$122,840.78
		9		\$138,027.72	\$6,338.18	\$4,723.75	\$5,442.17	\$63,338.50	\$111,801.14		\$126,200.03
		10		\$108,791.73	\$2,222.03	\$4,066.36	\$3,708.17	\$35,148.89	\$72,149.87	\$119.98	\$116,500.71
		11		\$120,089.19	\$3,938.70	\$5,287.06	\$4,848.64	\$39,948.29	\$91,942.10	\$193.22	\$117,975.57
		12		\$156,043.67	\$10,687.34	\$6,118.06	\$5,946.29	\$82,794.72	\$121,536.62		\$167,269.88
	2021	1		\$85,392.89	\$1,596.00	\$4,890.41	\$2,973.70	\$32,602.66	\$75,872.37		\$81,970.33
		2		\$76,355.66	\$4,073.20	\$3,263.65	\$2,794.50	\$13,683.24	\$67,735.04		\$63,002.08
							21.96				21.96
				\$1,968,985.34	\$95,661.10	\$38,490.03	\$96,197.87	\$735,506.50	\$1,658,717.80	\$884.23	\$1,589,243.80
											\$6,151,266.49

4. All other observations are written in the insights_log.

In my **insights_log** I'm going to note tab , what are the metrics and dimensions that I'm actually looking at, what is the key finding and who is the relevant team that is going to need this information.

Comparing the ranges of sales (or any KPI) across products (any other dimensions) can give me a sense for how the best performing segments compare to the worst performing segments.

In this case the best performing product can be **gaming monitors, nintendo switch and sony playstation** and the worst performing product can be **Gaming laptop, mouse and Gaming headsets**.

Next I can start looking into creating charts but before that I can add sparkline, which will give me an initial idea of trend of a metrics without needing a full graph, also will help me in deciding which graphs to use and why.

For Notable Segments / Business Context Part-

In order not to create many graphs and getting confused, thinking and gathering info again-

What is my main question? **How did total revenue dollars across all products performed during 2019 to 2022? Conduct initial analysis for managers in marketing , product and finance to understand high-level trends."**

what are my,

1. North star **Metrics?** **USD Price, Total Revenue**
2. North Star **Dimensions?** **Region, Time, Marketing Channels, Product**
3. North star Team Goals?

The marketing team wants to know: How can we get more customers or get them to buy more?

The finance team wants to know: Where is the revenue loss coming from?

Product team wants to know: what features or which product are doing good/ bad?

These north stars will help me determine how to slice the data, what insights I actually need to focus on and how I actually frame my recommendations to respective teams.

With that clarity changing my mindset from what happened? To Why happened?

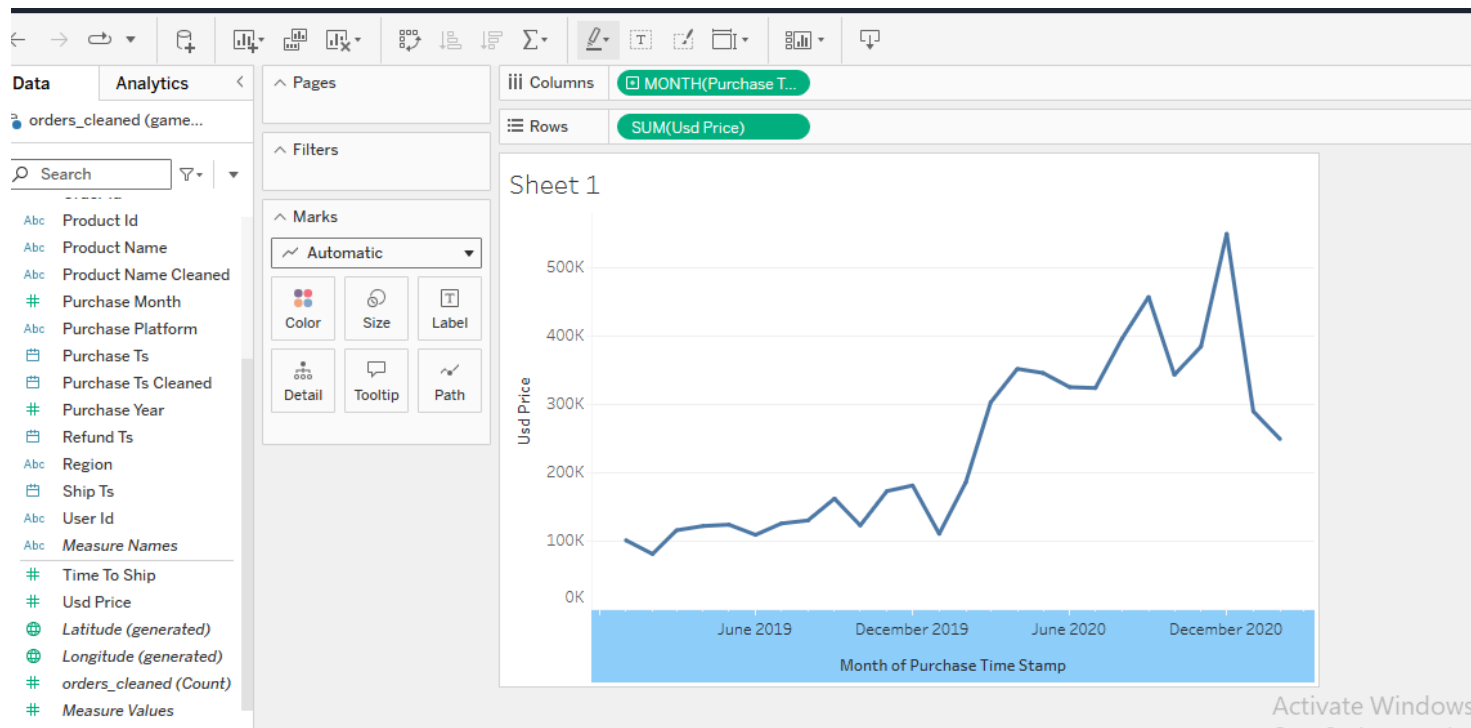
Moving my data into a **tableau** for better graph visualization, moving dimensions and structure of a graph.

With my main question of **What are overall trends in sales?** Line Graph of Sales over time seems to be a good option. Adding USD Price in columns and Purchase Timestamp in columns , checking on a monthly basis since I don't have much data, yearly won't give me any insights.

In overall USD price, I can see there are peaks at certain times of the year, so it's April 2020, Sep 2020 and Dec 2020. I'm also seeing there was a huge spike in Jan 2020, seems to be early COVID months then there was a dip in Jan 2021.

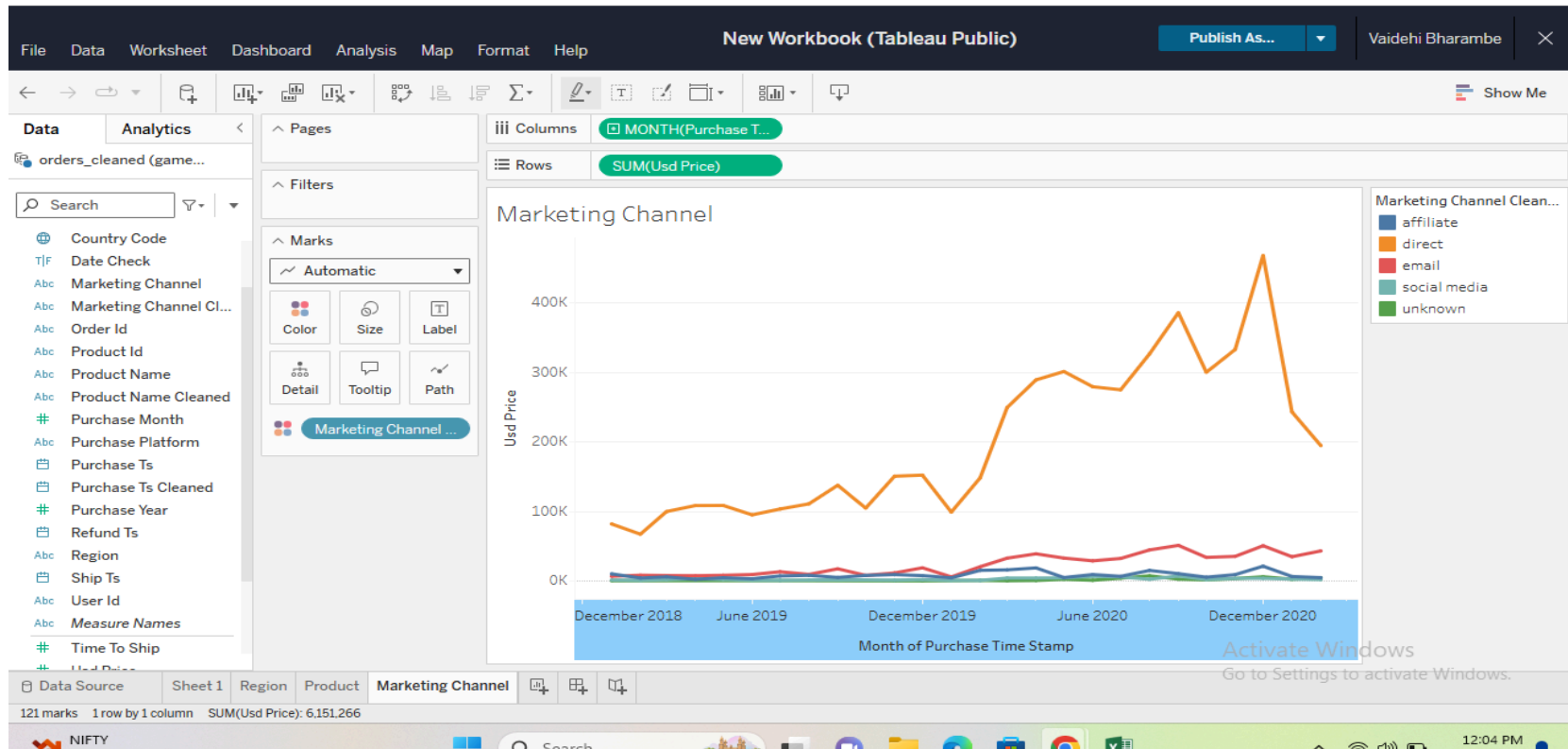
The overall trend is growing strongly, but there is a clear drop every December before it bounces back.

As an analyst, my next step would be to investigate why December drops and see how we can fix it.



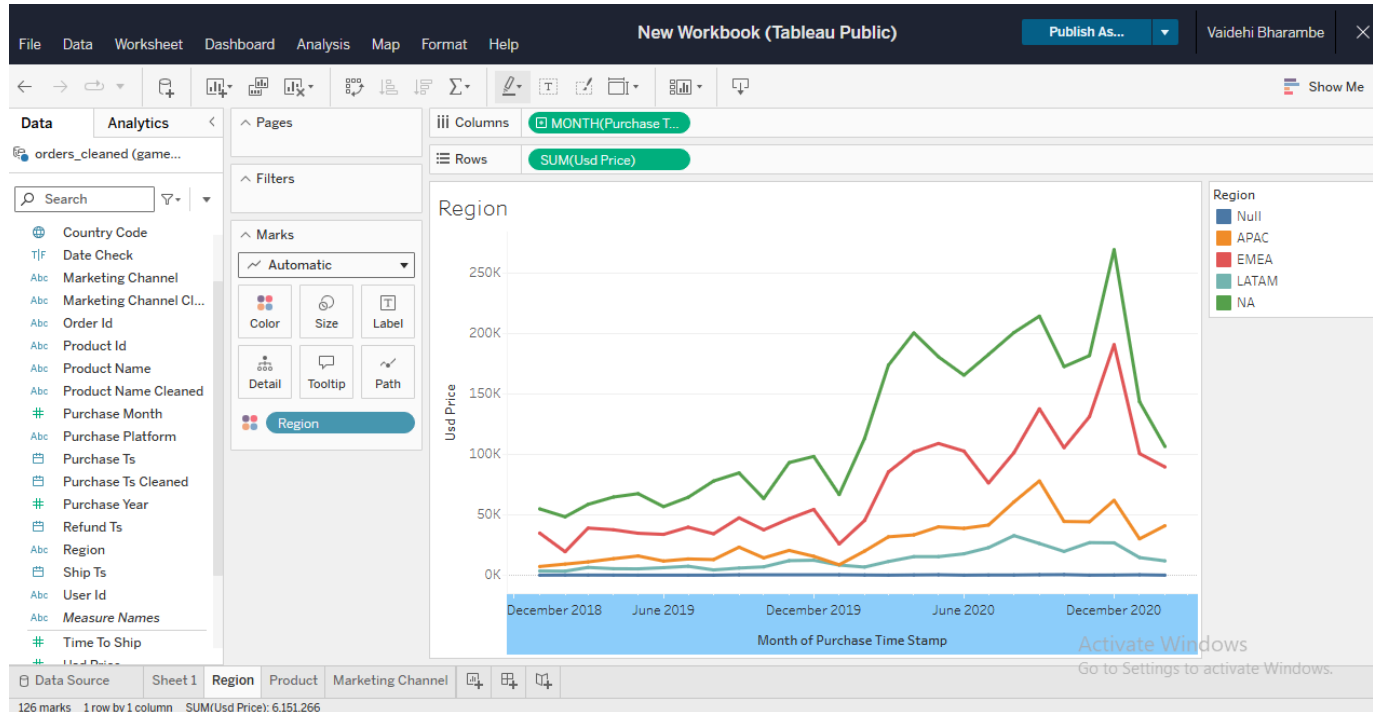
In order to do that, I am going to split this by three dimensions (**Region, Product and Marketing channel**), to see whether there is any specific dimension that is the cause of this dip. Duplicating the same graph and adding the three dimensions,

When I look at
1. USD sales by Marketing Channels,



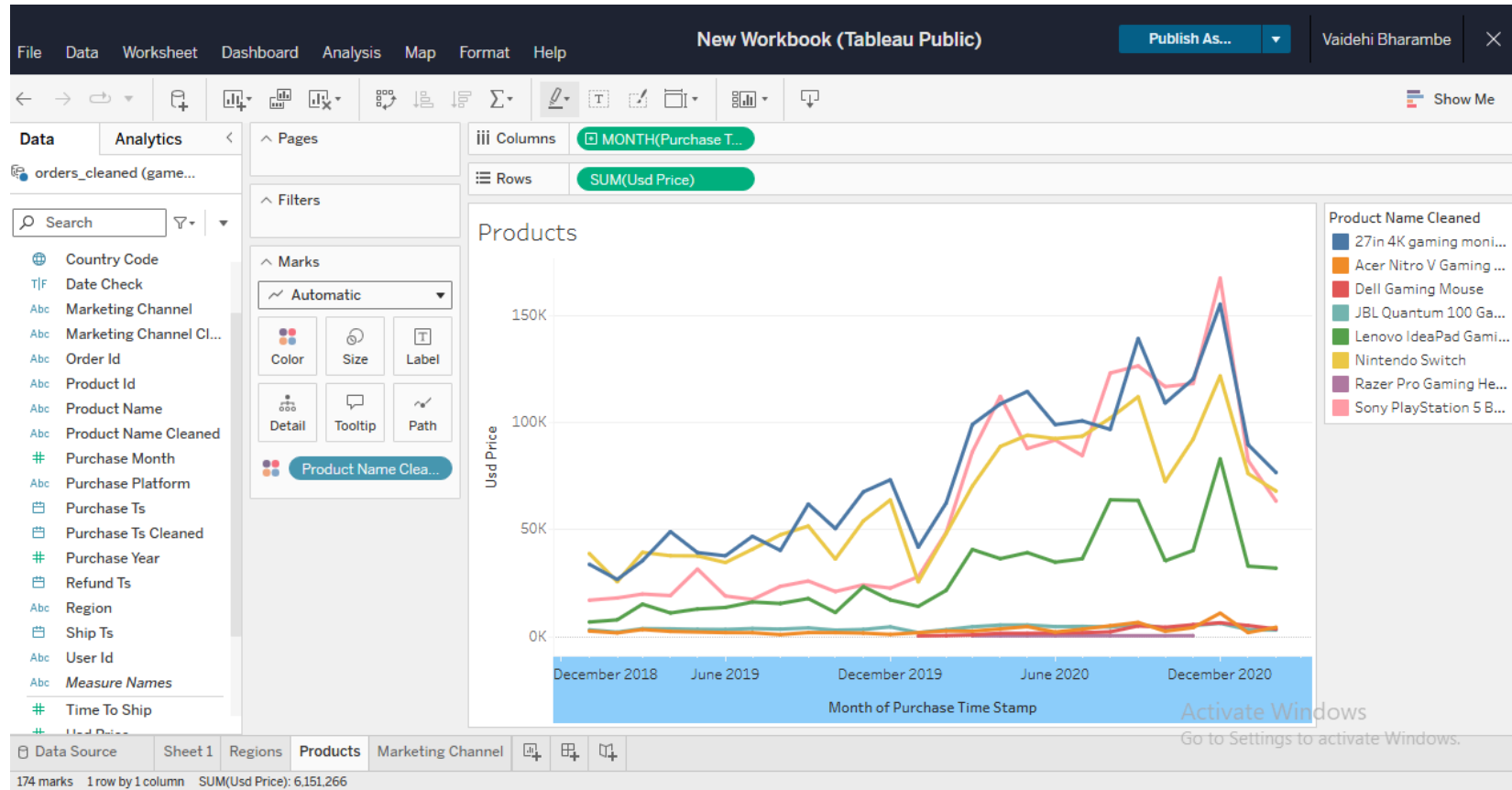
I can see that most of the revenue actually came from direct, i.e ppl came directly to the website and made purchases there, without clicking on any email or social media link. **So, in Marketing channels direct seems to be the main driver for big dips in Dec 2019 and Dec 2020.**

2. USD sales by Region,



I can see that it seems like a global trend because every single region is following the same pattern. The most steep is in **NA** i.e North America. Piecing things together - **North America and Direct channels are main drivers for the dips at the end of Dec 2019 and Dec 2020.**

3. USD sales by Products,



I can see that out of all the products there are 3 to 4 products that make up the majority of the revenue and three(**gaming monitor, nintendo switch,Lenovo ideapad gaming 3**) of them had a major dip. Whereas for **sony playstation**, I can see gradual increase in the sales. From my initial excel analysis, ignoring the headset and gaming laptop category because they are so low in revenue that I can ignore them.

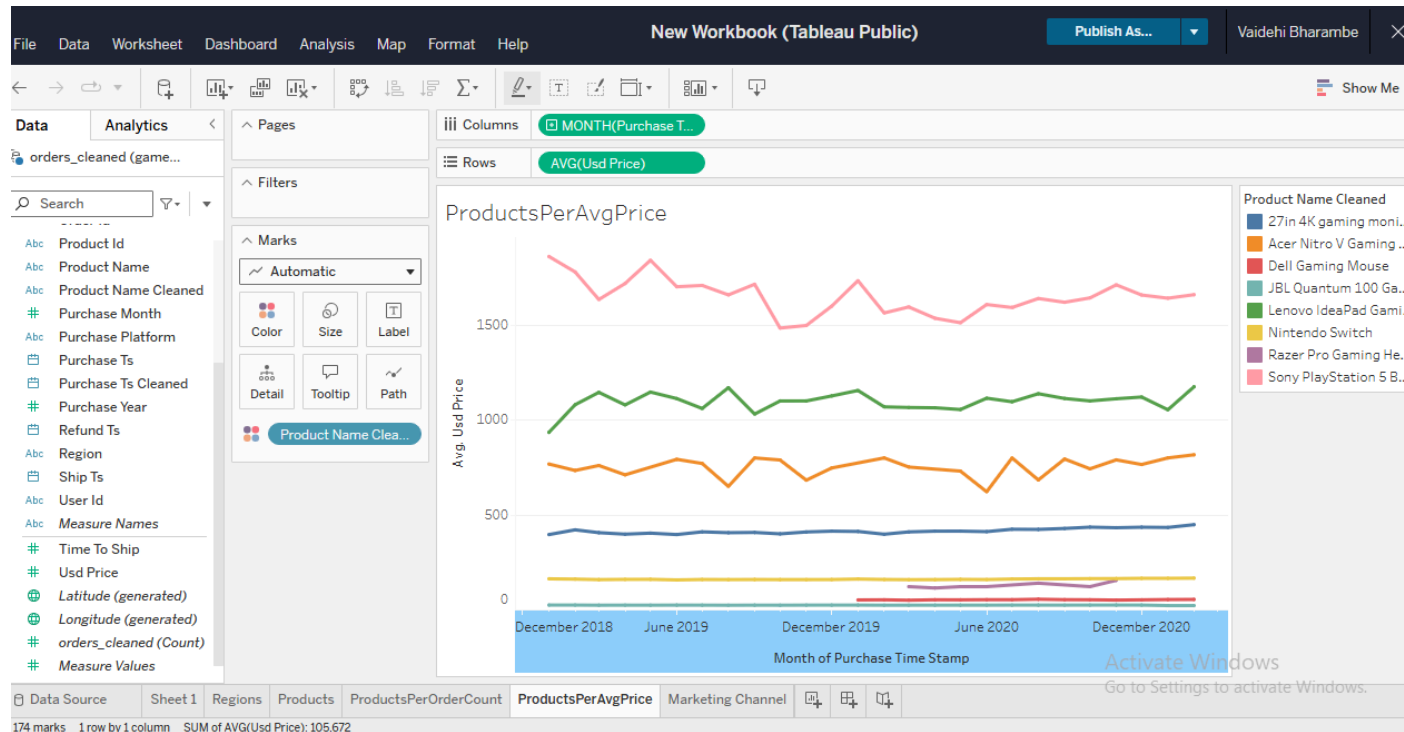
Although **Lenovo Ideapad Gaming 3** has some potential, like if we work on it we can increase its sales.

I can also see that out of my top 3 products the **pink line (Sony Playstation)** has a huge dip in early 2021, as compared to other 2 products, something that needs to be kept in mind for future references.

Using my domain knowledge **Exploring total count and average order value by product** here,
When I am saying Total Sales per product , it means total sales are dependent on other two metrics i.e total quantity * the total average price
$$\text{Total sales} = \text{Order Count} * \text{Avg Order Value}$$

I want to understand if it is the **revenue or is the USD price** fluctuating, meaning is it that the people are buying less or is it because people are buying the same amount but the price went down.
I have my total sales by product but now I want to look at the order count by product as well as the avg price by product.

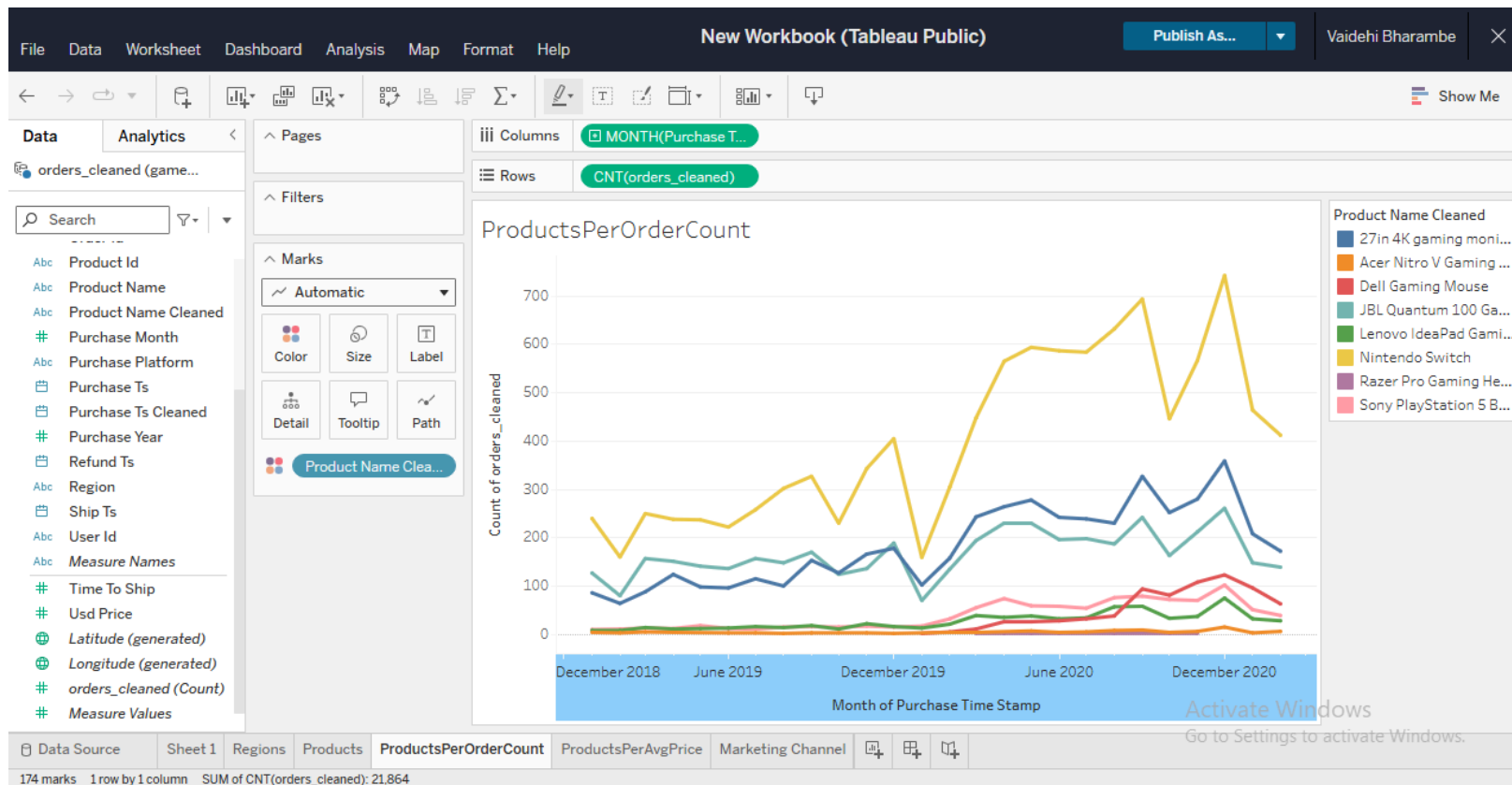
Mostly looking at the **Blue**, **yellow** and **pink** lines because those are products that are main drivers for the dip. Avg order value per product



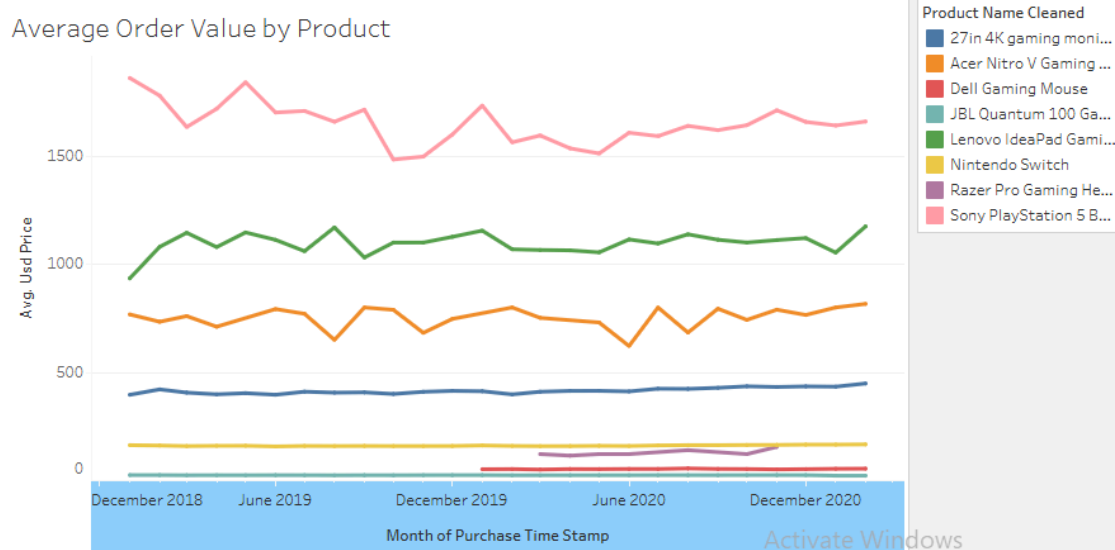
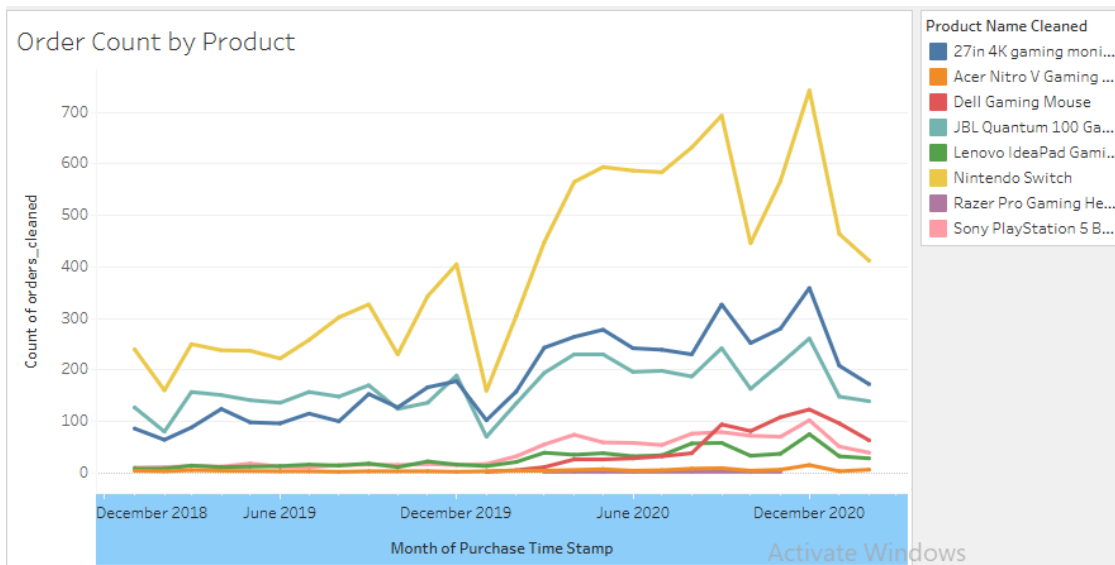
What's the story here? The Gaming monitor and Nintendo switch are quite stable here in AOV, whereas Sony playstation has more fluctuating prices, major dip in March 2019 then Oct 2019 and May 2020. Although stably increasing after Nov 2020.

From this graph I am not getting much info why AOV went down as there can be many possible reasons like USD dollar conversion rate may be different in different countries or there might be some products more popular in different countries, or can be because of discounts running for certain products.

So looking in order count now, remember looking at **Blue**, **yellow** and **pink** lines.



I can see for the Nintendo switch and Gaming monitor there are dips in the beginning of 2019 and 2020. For the pink line i.e for Playstation with highest AOV there seems to be a dip in April 2020 and then again in Jan 2021. Since **Playstation has the highest AOV** , i.e it's the most expensive product a strong dip or any fluctuation in order count is going to have an outsized impact on the total revenue.



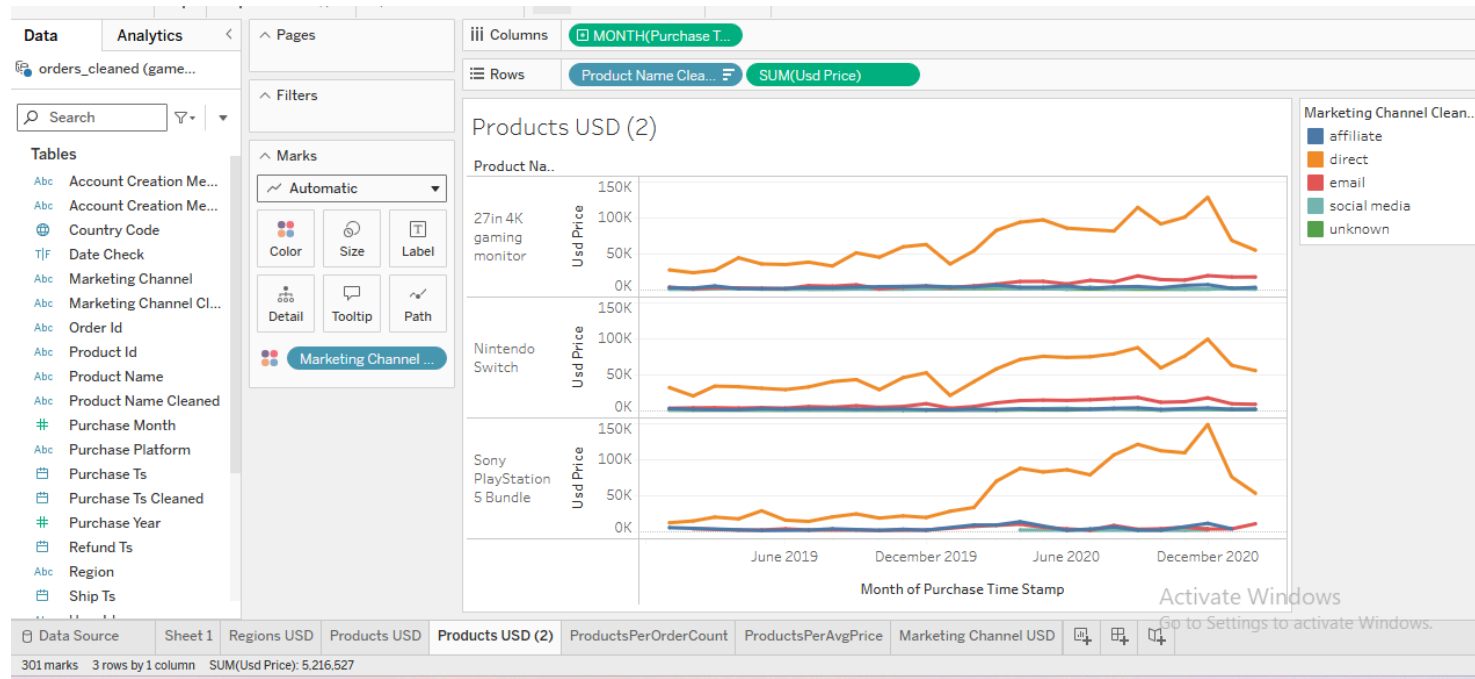
When I divide USD price into 2 different components order count and avg order value, Looking at them together, piecing the story here about why the total sales went down? I can see the order count for the most popular products went down

meaning people were buying less frequently and for **Sony Playstation** specifically the Average order value went down as well. Because the Average order value for Sony Playstation is the highest out of the top 3 products, then any kind of decrease in the order count can have a significant impact on the revenue.

Next Question , stakeholder may ask is **What was happening in those top three products that caused the sales dip?** So I am going to take these top three products and see if there are any specific regions or channels that might be affecting the sales.

Remember when I did the sanity check on Marketing channels in 1. USD sales by Marketing Channels, there was an outsized impact of **DIRECT** channels. Lets see what it looks like split by product. Duplicating the product USD tab.

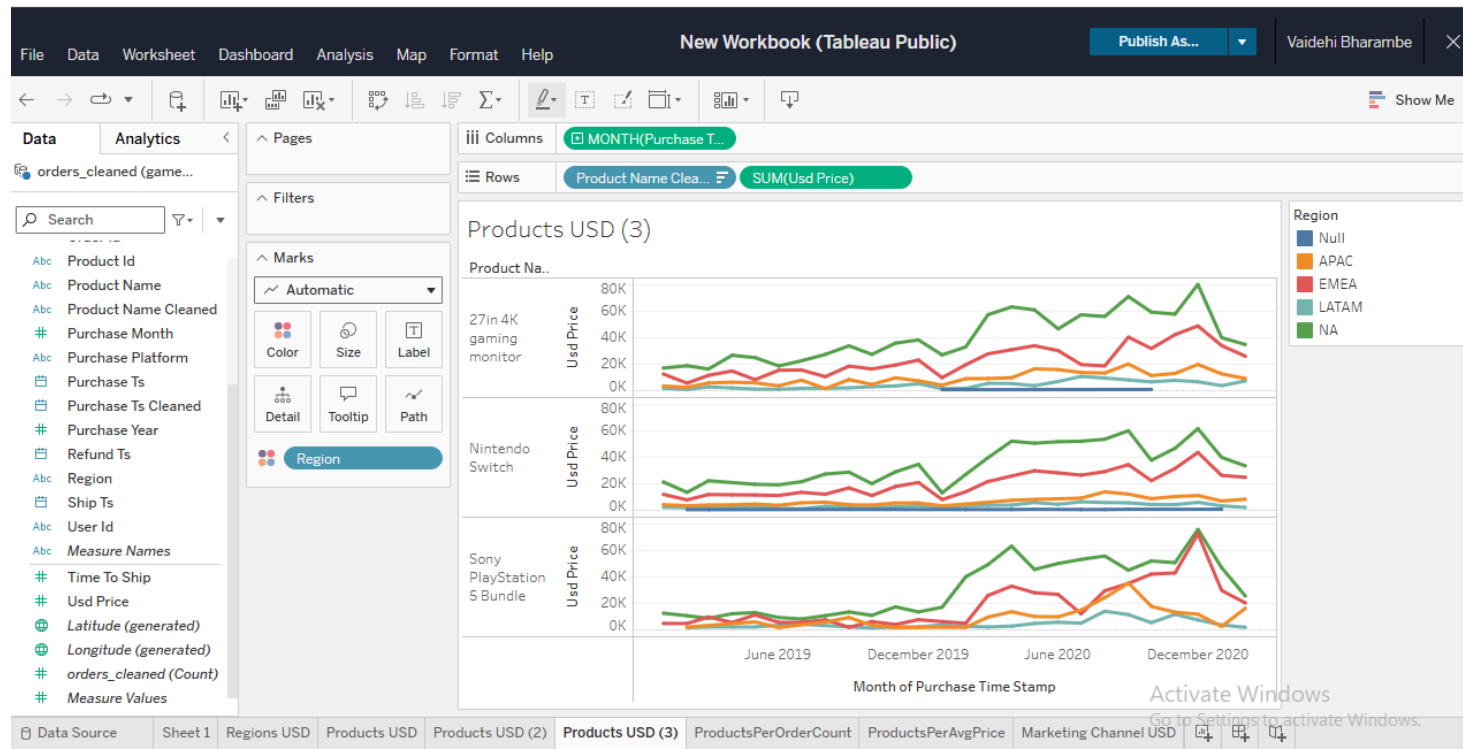
I am going to split the product USD line graph by different marketing channels, by moving product name to the rows and going to split out the series by marketing channels so that I can see within each product what was the split. Removing extra products and sorting the products by field > descending > USD price, so that I get the highest to lowest product.



Nothing much of information here, Gaming monitor and Nintendo switch are showing almost the same trend, Sony Playstation have a significant dip after Dec 2020. Direct can be out of Marketing teams reach but some of the possible reasons can be,

1. Competitors' popularity for the specific product.
2. Did we change the location of the product on the website?
3. Were there any promotions running at that time?

Going to do the same thing for Product split by Regions, trying to look for a one region that is responsible across these three products for the dips that we saw earlier.



I can see global ups and downs, whereas for Sony Playstation there is an oversized dip in North America and EMEA(Europe, the Middle East, and Africa) part.

So, one of the main insights I am coming up here is, "Direct NA and EMEA sales seems to be the biggest causes of the dip in sales at the end of Dec 2020.almost 72K to 30K "

My tableau viz for Gamezone can be found here:

<https://public.tableau.com/app/profile/vaidehi.bharambe/viz/GameZoneEDAAnalysis/overallUSD>

Next, going to note and document all my findings tableau tab wise in insights_logs for my future references.

Metrics Dimensions	USD_PRICE PURCHASE_PLATFORM, MARKETING_CHANNEL, ACCOUNT_CREATION_METHOD, PURCHASE_TS, SHIP_TS
SCAN FRAMEWORK	How did the total dollars revenue across the different products perform during 2019-2022. Conduct initial analysis to help product, marketing and finance managers understand the high level trends.
Stakeholder Goals	What dimensions and KPI's matters the most?
Columns and Coverage	What data do we have available and how can we use it?
Aggregates and Anomalies	What are the high level stats , outliers and unexpected patterns
Notable Segments	Slice by category , time or other key dimension to surface early insights
Tab	metric / dimension Findings Relevant team
pivot table 1	usd price / month Total sales across 2019- 2022 is ~6.1M, ranging in monthly sales from \$80K to \$500K finance team
pivot table 1	usd price / product Top performing product is gaming monitor (almost \$2M total sales), worst performing product is gaming headset(\$800K total sales), might be missing data, need to check. finance team, product team
pivot table 1	usd price / product category worst performing category is headset, which are responsible for less than 2% of total sales. product team
pivot table 1	usd price / month Dec 2020 saw a large spike in sales- seems like fall and winter months before the new year performs the best. May be related to holiday season or promotion at that time. marketing team
pivot table 1	usd price / product All products had a significant spike in sales 2020 onwards, following the general trend in total sales. This indicate companywise effect due to COVID. finance team
pivot table 1	usd price / product Almost all products exhibited a spike in Dec 2020 - look into promotions or campaigns that happened at this time marketing team
Deep Dive Analysis	
Tabela Tab	Deep Insight Recommendation
overall USD	sales more than doubled in early 2020, with all time highs in Sep and Dec 2020. Sales dropped significantly in early 2021. Do sanity check on key dimensions to find key drivers.
marketing USD	Direct is the main driver of sales- all other channels are weak in comparison. Investigate with marketing team on email and social media strategies.
region USD	All regions shows same dips in Dec 2019 and Dec 2020, indicating Global trend. Use marketing and product as main dimension to explore my analysis.
product USD	Gaming monitor, Nintendo Switch and Sony PlayStation are the main drivers, all shows the same dip pattern in early 2020. Look into these products by region and marketing channels.
products marketing	Out of top 4 products Gaming Monitor, Nintendo Switch, Lenovo IdeaPad Gaming shows dip in early 2020. For Gaming Monitor, there is an oversized dip in North America and EMEA (Europe, the Middle East, and Africa) part. Investigate Sony PlayStation in NA - did the location on the website change, was there any competitor?
product region	Global ups and downs, whereas for Sony PlayStation there is an oversized dip in North America and EMEA (Europe, the Middle East, and Africa) part.

Business Recommendations:

1. Finance Team Insight: The Core Revenue Drivers & Dips

- **The Finding:** Total revenue reached \$6.1 million, heavily concentrated in just three top-tier products: Gaming Monitors, Nintendo Switch, and Sony PlayStation.
- **The Deep Dive:** The dramatic sales drop-off at the end of December 2020 (a decline from ~\$72k to \$30k) was not a single-product issue; it was a systemic global trend driven primarily by the North American (NA) and EMEA markets.
- **The "Why":** Breaking total sales into its core components revealed that the dip was caused by a drop in **Order Count** (fewer transactions), combined with a simultaneous drop in **Average Order Value (AOV)** specifically for the high-margin Sony PlayStation line. Because the PlayStation commands the highest price point, any drop in its volume disproportionately damages the company's bottom line.

2. Marketing Team Insight: Over-Reliance on Direct Traffic

- **The Finding:** The vast majority of revenue is driven by "Direct" website traffic, while paid or organic acquisition channels (Email, Social Media) remain flat and low.
- **The Recommendation:** Because Direct traffic is highly volatile and tied to organic seasonality, Marketing needs to build a defensive buffer. We recommend launching targeted re-engagement email campaigns and social media promotions specifically timed for late December and early January to flatten out the post-holiday seasonal cliff.

3. Product Team Insight: Portfolio Health & Inventory Optimization

- **The Finding:** Products like the Razer Pro Gaming Headset and various gaming laptops are generating negligible revenue and can be deprioritized from main web listings. Conversely, the Lenovo Ideapad Gaming 3 shows upward potential.
- **The Recommendation:** Investigate the early 2021 Sony PlayStation dip. Was this caused by supply chain shortages (extremely common for consoles in 2021), a change in website placement, or a lack of promotional bundles? Ensure inventory for high-AOV items is optimized ahead of Q4 to avoid leaving money on the table.

I made a few line graphs, focusing on clarity and simplicity over complex charts and graphs to look at the trends from COVID because I wanted to explore how a macro event could actually affect an e-commerce company.

I understand EDA is an iterative process with a lot of back and forth between stakeholders but since there were no stakeholders involved in my portfolio project I kept it simple focusing on clarity of overall structure and framework that I can use on my job.

Next will be How to convert these found insights into Visualizations.